



UNIVERSITY of
RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY
SCHOOL OF ICT

DEPARTMENT OF Computer Science

The campus where the module exam is conducted: NYARUGENGE & GAKO

END OF SEMESTER EXAMINATION - ACADEMIC YEAR 2024-2025

YEAR:1 SEMESTER: II

MODULE CODE and NAME: CSC1261 Computer Programming in C

DATE: 16 June, 2025

TIME: 2 hours

MAXIMUM MARKS = 50

INSTRUCTIONS

1. This paper contains **FOUR (4)** questions.
2. Answer **THREE (3)** Questions only:
Question ONE (1) from Section "A" is Compulsory and Answer any TWO (2) Questions from Section "B"
3. No written materials allowed.
4. Do not forget to write your Registration Number.
5. Write all your answers in the booklet provided
6. Do not write any answers on this questions paper.
7. Start each question in a **NEW** page

MODULE CODE & TITLE: CSC 1261 COMPUTER PROGRAMMING in C

SECTION A

COMPULSORY QUESTION

Question 1 (20 marks)

- 1.1 What do you mean by pointer? List 3 items required to define it. (2 marks)
- 1.2 By using examples, explain the 2 types of languages. (2 marks)
- 1.3 Differentiate structure from array. (2 marks)
- 1.4 List all storage classes used in C Programming . (2 marks)
- 1.5 What are the advantages of using functions in C Programming. (2 marks)
- 1.6 Write a C program for computing the average of 3 integers (3 marks)
- 1.7 Make a flowchart to divide any two numbers (3 marks)
- 1.8 What do you mean by string.h? Give 2 functions defined under string.h (2 marks)
- 1.9 What is the error difficult to handle? Motivate your answer. (2 marks)

SECTION B

ANSWER ANY TWO QUESTIONS

Question 2 (15 marks)

- a) With a help of flowchart, differentiate while loop from do while loop. (3 marks)
- b) By using while loop, write a C program that prompts a user to type in an integer number n and check whether it is even or odd. (4 marks)
- c) Find and comment on errors in the piece of code below: (4 marks)

```
#include<stdio.h>
main()
{
    int number;
    int *ptr;
    ptr=number;
    *ptr=&number;
    ptr=&number
    *ptr=number,
}
```

- d) Write a C program to convert the RWF (Rwandan Francs) into USD (United State Dollars) and vice versa (1USD=1,450 RWF) (4 marks)

Question 3 (15 marks)

- a) Make an application that calculates $y = \frac{\sqrt{x^3-4}}{x^2}$ (4 marks)
- b) Identify errors in the below piece of code. (3 marks)

```
main()
{
    int arr[5];
    arr[1]=8.5;
    arr[5]=10;
}
```

- c) Write a C program that prompts a user type in integer number n and displays a multiplication table of n using for loop. (4 marks)

d) Write a C program that:

1. Defines a structure called ~~Student~~^{Employee} with the following members:

int id; char name[20] and float salary

(1mark)

2. Declares an array of 3 Employee records.

(1mark)

Uses a for loop and scanf to read employee data from the user.

(1mark)

Uses another for loop to display the employee details.

(1mark)

Question 4 (15 marks)

- a) Locate the error(s) in the below structure definition.

(3 marks)

```
Struct student
{
    int rolno=880;
    float marks;
    char rollno;
}
```

- b) Write a program that will prompt a user to enter marks of students between 0 and 100 and make decision as follow :

100-80: Upper Division

79.9-60: Lower Division

59.9-50: Pass

<50: Retake

(5 marks)

- c) Explain all types of functions.

(4 marks)

- d) Write a program to display the following:

(3 Marks)

```
1*1=1
2*2=4
3*3=9
4*4=16
```

GOOD LUCK

